

CECS Risk Identification + Risk Register (Part 1 to 3)

ID	Category	Risk	Likelihood	Impact	Risk	Priority	Mitigation (before)	Contingency (after)	Trigger/Signal	Owner	Status
R1	Technical	Filter Lag: multiple concurrent filters slow queries/feed updates	4	4	16	High	Index/filter queries; cache base event list; debounce UI; perf tests in CI	Disable costly combos; degrade gracefully (feed-only / no map refresh); rollback last filter change	Filter response >2s / timeouts / DB CPU spike	Requirements and Design Lead / Backend	Open
R2	Technical	Real-Time UI Breakage: AJAX/live updates break runtime or inconsistent UI state	3	4	12	Medium	Incremental integration; feature flags; unit+integration tests for async calls	Turn off live updates; fall back to manual refresh; hotfix to last stable build	JS errors; feed stops updating; inconsistent state reports	Frontend	Open
R3	Technical	Map Integration Failure: feed/filter changes break map rendering/pins	4	4	16	High	Separate MapService from filtering; contract tests for pin payload; regression tests	Fallback to external map link only; hide embedded map temporarily	Pins missing/wrong; map crashes after filter change	Implementation and Integration Lead / Frontend	Open
R4	Technical	Injection Vulnerability : untrusted inputs allow SQL/script injection	4	5	20	High	Input validation; parameterized queries; output encoding; security review	Rotate credentials; restore backup; incident response + patch	Suspicious query patterns; unexpected DB writes	Requirements and Design Lead / Security/Backend	Open
R5	Technical	Stale Data: session cache shows outdated events after mid-session approvals	3	3	9	Low	Cache TTL; invalidate cache on approve/publish; show "last updated"	Force refresh prompt; reload on key actions	Mismatch between admin view and student view	Backend	Open
R6	Project Mgmt	Scope Creep on Maps: pin	4	4	16	High	Freeze scope;	Cut routing;	Requests for turn-by-	Project Lead	Open

		drop expands to full routing/navigation					define "map v1" = pin + external directions; backlog routing	ship minimal map; renegotiate scope	turn/ETA/route options		
R7	Project Mgmt	Requirement Volatility: cultural credit rules change late → redesign	4	5	20	High	Make eligibility rules configurable; flexible schema; confirm policy early	Change request process; re-baseline; drop non-essentials	Stakeholder policy update or new constraints	Project Lead	Open
R8	Project Mgmt	Big-Bang Integration: merge feed+filters+map at end → hard failures	4	4	16	High	CI from day 1; integrate small slices; regression suite	Rollback merges; isolate via feature flags; freeze new features	Huge merge PRs; unstable build; many failing tests	Project Lead	Open
R9	Team	Design Pattern Overhead: Strategy pattern bypassed → God Class returns	3	4	12	Medium	Short design doc; starter template; pair on first strategy; code review rule	Refactor sprint; architecture gate in PR checks	PR adds big if/else blocks inside EventService	Integration Lead	Open
R10	Team	Merge Conflicts: parallel refactor + security work causes Git conflicts	4	3	12	Medium	Branch plan; smaller PRs; frequent sync; module ownership	Conflict-resolution session; code freeze; cherry-pick clean commits	Repeated conflicts on same files; blocked merges	Integration Lead	Open
R11	Team	Test Decay: CI regression tests stop being maintained	3	4	12	Medium	Require tests in PR; treat tests as deliverable; keep critical-path suite	Rebuild minimal regression suite; remove flaky tests; re-stabilize CI	CI red for days; tests ignored/disabled	QA/All	Open
R12	External	Authentication Changes: university SSO endpoints	3	4	12	Medium	Abstract auth layer; mock adapter; avoid	Switch to mock auth for prototype;	Login failures spike; upstream auth errors	Backend	Open

		change/break login					hardcoding endpoints; monitor	manual accounts			
R13	External	Third-Party API Limits: map provider changes pricing/rules	3	3	9	Low	Use basic directions link format; avoid paid features; document fallback	Switch provider ; disable embedded map; show text location + link	Quota exceeded / billing warning / API errors	Project Lead/Frontend	Open
R14	User/Stakeholder	Data Inconsistency: free-text tags vary → filters inaccurate	4	4	16	High	Controlled tag dropdown; tag validation; staff guidance	Lock to whitelist ; cleanup script to normalize tags	Many near-duplicate tags; filter results feel random	Product/Frontend	Open
R15	User/Stakeholder	Privilege Escalation: students bypass access control to approve events/credits	4	5	20	High	Role Based Access Control server-side checks on every privileged request; least privilege; audit logs	Revoke sessions ; revert approvals ; restore records; patch + incident report	Unauthorized approvals; role mismatch in logs	Implementation and Integration Lead / Security/Backend	Open

Top priorities

- R4 (Technical/ mason) — 20: Injection Vulnerability: untrusted inputs allow SQL/script injection
- R7 (Project Mgmt/ kamal) — 20: Requirement Volatility: cultural credit rules change late → redesign
- R15 (User/Stakeholder/ solomon) — 20: Privilege Escalation: students bypass access control to approve events/credits
- R1 (Technical/ mason) — 16: Filter Lag: multiple concurrent filters slow queries/feed updates
- R3 (Technical/ solomon) — 16: Map Integration Failure: feed/filter changes break map rendering/pins
- R6 (Project Mgmt/ kamal) — 16: Scope Creep on Maps: pin drop expands to full routing/navigation